NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

METAL FORMING PROCESS

ADVANCE MANUFACTURING PROCESS (IM-307)

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1. INTRODUCTION TO METAL FORMING

1.1 METAL FORMING:

Metal forming is a fundamental manufacturing process that involves shaping metal workpieces through plastic deformation under the influence of external forces. Unlike machining, which removes material, or casting, which involves melting and solidifying metal, metal forming reshapes the material while maintaining its mass. This process often improves the material's mechanical properties, such as strength and toughness.

Metal forming is extensively used across numerous industries including automotive, aerospace, construction, and consumer goods due to its capability to produce high-strength, complex parts with excellent surface finishes and material utilization.

1.2 Fundamentals of Metal Forming

Principle of Plastic Deformation

The core principle of metal forming is plastic deformation. When a force exceeding the yield strength of a metal is applied—yet remains below its fracture point—the metal undergoes a permanent shape change without cracking.

Role of Stress and Strain:

- **Stress** is the internal force per unit area within a material caused by externally applied forces.
- **Strain** is the measure of deformation or displacement that occurs due to stress.

In **metal forming processes**, a controlled amount of stress is applied to cause **plastic strain**, allowing the metal to take a desired shape without cracking or failing. The material flows plastically under compressive, tensile, or shear stresses, depending on the process (e.g., rolling, forging, drawing).

The metal's **yield strength**, **strain hardening**, and **temperature** play key roles in how it deforms and how much force is required.

2. CLASSIFICATION OF METAL FORMING PROCESSES:

Metal forming techniques are categorized based on the type of stress applied and the working temperature:

Classification by Type of Stress

- **Compressive Forming:** Forging, Rolling, Extrusion
- **Tensile Forming:** Stretching, Expanding
- **Shear Forming:** Shearing, Blanking, Piercing
- **Combined Methods:** Deep Drawing, Bending

Classification by Working Temperature

- **Cold Working:** Performed at room temperature. Results in high strength and good surface finish but requires high force.
- **Warm Working:** Conducted at moderately elevated temperatures. Offers a balance between cold and hot working.
- **Hot Working:** Done above the metal’s recrystallization temperature. Allows large deformations with less force but offers lower dimensional accuracy.

Forging

- **Description:** Deformation of metal between dies using compressive forces
- **Types:**
  - Open-die forging
  - Closed-die (impression-die) forging
  - Press forging
- **Working Conditions:** Cold, warm, or hot
- **Applications:** Crankshafts, gears, connecting rods
- **Materials Used:**
  - Steel and alloy steel
  - Aluminum and its alloys
  - Titanium alloys
  - Copper and brass

1. Rolling

- **Description:** Metal is passed through rollers to reduce thickness or change cross-section
- **Types:**
  - Hot rolling
  - Cold rolling
  - Flat rolling
  - Shape rolling
- **Applications:** Sheet metal, I-beams, rails
- **Materials Used:**
  - Stainless steel
  - Aluminum
  - Copper

2. Extrusion

- **Description:** Metal is forced through a die to produce objects with a fixed cross-sectional profile
- **Types:**
  - Direct (forward) extrusion
  - Indirect (backward) extrusion
  - Hot and cold extrusion
- **Applications:** Tubes, rods, structural profiles
- **Materials Used:**
  - Aluminum and its alloys
  - Copper
  - Magnesium
  - Lead
  - Steel (hot extrusion)

3. Drawing

- **Description:** Metal is pulled through a die to reduce cross-sectional area and increase length
- **Types:**
  - Wire drawing
  - Tube drawing
  - Bar drawing
- **Working Conditions:** Cold or hot
- **Applications:** Wires, tubes, rods
- **Materials Used:**
  - Steel (carbon and stainless)
  - Aluminum
  - Copper
  - Brass

5. Sheet metal Forming process

a. Stretch Forming

- **Description:** Sheet is stretched and bent over a die.
- **Applications:** Aircraft panels, curved components
- **Limitations:** Shallow contours only
- **Materials Used:** Aluminum and its alloys, Copper

b. stretching

- **Description:** Diameter of a tube or hollow shape is increased.
- **Applications:** Pipe fittings, nozzles
- **Material Used:** Steel (carbon and stainless), brass

c. Deep Drawing

- **Description:** Sheet is pulled into a die to create deep, hollow parts
- **Applications:** Cans, sinks
- **Challenges:** Wrinkling, tearing
- **Material used:** Mild steel, Stainless steel, Aluminum

Metal forming is an essential process in modern manufacturing, offering efficient, high-strength solutions for producing a vast array of metal components. From basic bending to complex hydroforming and coining, each method serves specific needs across various industries. Understanding the fundamentals, process classifications, and practical considerations is critical for optimizing production and ensuring quality outcomes in formed metal products.

3. DETAILED CASE STUDY:

3.1 WHAT IS FORGING?

Forging is a metalworking process in which a piece of metal is shaped by applying **compressive forces**. These forces are usually delivered by a **hammer**, a **press**, or a **die** (a custom-shaped tool). Forging can be performed at various temperatures depending on the type of metal and the desired properties of the final product.

The main goal of forging is to produce components that are **stronger**, **more durable**, and **resistant to fatigue or stress** than those made using other manufacturing processes like casting or machining

How Forging Works

In forging, metal is **deformed plastically**, meaning it changes shape without cracking or breaking. This is achieved by pushing the atoms in the metal into new positions, which aligns the **grain structure** of the material. This grain flow is crucial because it increases the strength of the final part, especially in the direction of the applied force.

Key Characteristics of Forging

- **Grain Refinement**: The grain structure of the metal becomes finer and more aligned, which improves its mechanical properties.
- **High Strength**: Forged parts are stronger than cast or machined parts due to grain flow and work hardening.
- **Durability**: Forged components are highly resistant to impact and fatigue, making them ideal for heavy-duty applications.

Why Forging is Used

Forging is widely used in industries where the strength and reliability of parts are critical. For example:

- In the **automotive industry**, forged crankshafts and connecting rods withstand high stress.
- In **aerospace**, forged components are used for landing gear and turbine parts.
- In **oil & gas**, heavy-duty forged valves and flanges are used under extreme conditions.
- More **economical** for mass production once the tooling is in place

Limitations of Forging

- High **initial cost** for dies and equipment
- Not suitable for very **complex shapes**
- Requires **skilled labor** and careful temperature control

In summary, **forging is a reliable and powerful process** for manufacturing high-strength, high-performance metal parts by shaping metal through compressive force. It has been used for centuries—from blacksmiths making swords to modern factories producing aircraft components.

3.2 FORGING PROCESS: STEP-BY-STEP EXPLANATION:

(PRE-PROCESSING)

1. Material Selection

Before forging begins, the right type of **metal or alloy** is selected based on the application.

- Common materials: **Steel, aluminum, titanium, copper**
- Considerations: **Strength, toughness, ductility, corrosion resistance**

2. Heating the Workpiece

In **hot** and **warm forging**, the metal must be heated to make it easier to shape.

- **Hot Forging**: Above **recrystallization temperature** (e.g., 1100–1250°C for steel)
- **Warm Forging**: Between **650–850°C**
- **Cold Forging**: No heating required (done at room temperature)

Why heat? Heating softens the metal, reduces force required, and helps avoid cracking.

3. Pre-forming the Metal (Optional Pre-processing)

Sometimes, the heated metal is roughly shaped or cut into a size/shape called a **billet**, **slug**, or **preform** to make the forging process smoother and more efficient.

(MAIN PROCESS)

4. Forging

This is the **core step** where the metal is shaped using **compressive force** applied by a:

- **Hammer**
- **Press**
- **Roller**
- **Die**

Types of Forging:

- **Open-Die Forging**: Metal is pressed between flat dies—used for large, simple parts.
- **Closed-Die Forging** (Impression Die): Metal is shaped inside a mold—used for complex shapes.
- **Seamless Ring Rolling**: A ring is forged from a donut-shaped blank and expanded.

Outcome: The metal takes the shape of the die and becomes denser and stronger.

5. Flash Removal (Trimming)

In **closed-die forging**, extra metal (called **flash**) may be squeezed out of the die cavity. This is **trimmed off** using a press or machining tool.

(POST-PROCESSING)

6. Heat Treatment

After forging, heat treatment may be done to improve mechanical properties:

- **Annealing** – Relieves internal stresses
- **Quenching and Tempering** – Increases hardness and toughness
- **Normalizing** – Refines grain structure

7. Cleaning & Surface Finishing

The forged part may undergo:

- **Shot blasting** or **pickling** to remove scale and oxide
- **Grinding or machining** to improve surface finish or meet exact dimensions

8. Inspection & Quality Control

Parts are inspected to ensure they meet quality standards:

- **Dimensional checks**
- **Ultrasonic testing** or **X-ray** for internal flaws
- **Hardness and tensile testing**

9. Final Machining (If required)

Some forged parts need **final machining** (e.g., drilling, turning, milling) to meet precise tolerances.

10. Assembly or Shipping

After inspection and finishing, the parts are either:

- **Assembled** into a final product
- **Shipped** to customers or manufacturers

3.3 EQUIPMENT AND TOOLS USED IN FORGING:

Forging relies on a combination of **heavy machinery**, **dies**, and **support tools** to shape metal under high pressure. The type of equipment used depends on the **type of forging** (hot, cold, warm), **size of the workpiece**, and **precision required**.

1. Forging Presses

➤ **Hydraulic Press**

- **How it works**: Uses hydraulic pressure to apply slow, controlled force.
- **Advantages**: Precise and good for large or intricate parts.
- **Used for**: Closed-die forging, open-die forging, warm forging.

➤ **Mechanical Press**

- **How it works**: Uses a crankshaft, eccentric cam, or other mechanical systems to drive the ram.
- **Advantages**: High-speed forging, suitable for mass production.
- **Used for**: Cold forging, shallow parts.

➤ **Screw Press**

- **How it works**: Converts rotary motion into linear motion using a screw mechanism.
- **Advantages**: Offers a balance between speed and control.
- **Used for**: Precision parts with moderate deformation.

2. Forging Hammers

➤ **Drop Hammer**

- **How it works**: A heavy hammer is dropped onto the workpiece placed on the die.
- **Advantages**: Simple, fast, and good for repeated blows.
- **Used for**: Open-die forging, blacksmithing, and small parts.

➤ **Power Hammer**

- **How it works**: Uses steam or air pressure to lift and drop the hammer repeatedly.
- **Advantages**: More control and power than a manual hammer.
- **Used for**: Small to medium-size forgings.

3. Forging Dies

Dies are the **most critical tools** in forging. They determine the shape and accuracy of the final product.

➤ **Open-Die (Flat Die)**

- No cavity in the die; metal is deformed freely between flat or slightly contoured dies.
- Used in **open-die forging** for large parts.

➤ **Closed-Die (Impression Die)**

- The die has a cavity that matches the shape of the desired part.
- Metal flows inside the cavity and takes the exact shape.
- Used for **complex and high-volume parts**.

➤ **Flash Trimming Dies**

- Used to **remove excess metal (flash)** after closed-die forging.

4. Heating Equipment

Forging often starts by heating the metal to make it easier to deform.

➤ **Furnaces**

- Types: Gas-fired, electric, induction
- Used for: Heating billets to forging temperature (especially in hot and warm forging)

➤ **Induction Heaters**

- Use electromagnetic induction to heat metal quickly and efficiently.
- Common in **automated or cold forging lines** for pre-heating.

5. Handling and Support Tools

➤ **Tongs**

- Used to hold and manipulate hot workpieces manually in smaller forging setups.

➤ **Manipulators**

- Robotic arms or powered tools for handling large and hot forgings safely and accurately.

➤ **Lubrication Systems**

- Apply lubricants to dies or parts to reduce wear and friction.

6. Inspection and Measuring Tools

- **Calipers and Micrometers** – For dimensional accuracy.
- **Ultrasonic Testing Equipment** – For internal defect detection.
- **Hardness Testers** – To verify material properties.
- **Coordinate Measuring Machines (CMMs)** – For precise geometry checks.

3.4 ADVANTAGESOF FORGING

1. **High Strength and Toughness**: Forged parts have superior mechanical properties due to grain refinement and alignment, making them very strong and impact-resistant.
2. **Better Fatigue and Stress Resistance**: The continuous grain flow achieved during forging increases resistance to fatigue, shock, and vibration.
3. **Improved Structural Integrity**: Forging reduces internal defects such as porosity, shrinkage cavities, or inclusions, making the parts more reliable.
4. **Efficient Material Usage**: Forging has relatively low material wastage compared to machining or casting, especially with near-net shape forging.
5. **High Production Rates**: Once the dies and tools are set up, forging allows for **mass production** with consistent quality and speed.

3.5 DISADVANTAGES OF FORGING

1. **High Initial Cost**: The cost of dies, molds, and forging equipment is high, making it uneconomical for small production runs.
2. **Limited to Simple or Moderately Complex Shapes**: Forging is not ideal for highly complex geometries or hollow/thin-walled structures.
3. **Die Wear and Maintenance**: Forging dies are subjected to extreme conditions and wear out quickly, requiring regular maintenance or replacement.
4. **Requires Skilled Labor and Control**: Forging demands proper control of temperature, pressure, and timing, and requires trained personnel to operate safely and effectively.
5. **Post-Processing Often Required**: Additional steps like **heat treatment**, **machining**, and **surface finishing** are usually needed to achieve final specifications and tolerances.

4. MECHANICAL PROPERTIES & MATERIAL BEHAVIOR IN METAL FORMING

4.1 INFLUENCE OF METAL-FORMING PROCESSES ON MECHANICAL PROPERTIES:

Metal forming is more than just shaping — it directly influences the internal structure (microstructure) and mechanical properties of metals. As the metal is plastically deformed, it undergoes changes that affect its strength, hardness, ductility, and toughness.

Plastic Deformation & Property Change:

When a metal is subjected to forces beyond its elastic limit, plastic deformation occurs. During this, the grain structure changes, dislocations multiply, and the metal's properties are modified.

Process-Specific Influence

Metal-Forming Process	Mechanical Property Impact
Rolling	Grain elongation; increased strength along rolling direction (anisotropy).
Forging	Refines grain structure; improves strength and toughness.
Extrusion	Develops uniform grain flow; enhances directional strength.
Drawing	Causes work hardening; improves tensile strength but reduces ductility.
Stamping/Bending	Induces residual stresses; may lower fatigue strength if not relieved.

4.2 KEY CONCEPTS IN METAL FORMING

1. Work Hardening (Strain Hardening): Work hardening is the phenomenon where a metal becomes stronger and harder as it is plastically deformed.

Mechanism:

- During deformation, dislocations (defects in the crystal lattice) are introduced.
- As more dislocations pile up, it becomes increasingly harder for further movement, increasing strength.

Effects:

- Increased yield strength.
- Increased hardness.
- Decreased ductility.

2. Anisotropy: Anisotropy refers to the directional dependence of a material's mechanical properties.

Causes in Metal Forming:

- Metal-forming processes, especially rolling, stretch and align grains in one direction.
- This alignment makes the properties (like strength and ductility) vary depending on the direction of the load.

Implications:

- In rolled sheets, the tensile strength is usually higher along the rolling direction than across it.
- Engineers must consider this when designing components, especially in automotive, aerospace, and structural applications.

3. Residual Stresses: Residual stresses are locked-in stresses that remain in a material after the external force has been removed.

Causes in Metal Forming:

- Non-uniform deformation.
- Uneven cooling rates (e.g., in hot forming or welding).
- Bending and stretching in stamping or rolling.

Impact:

- Can cause dimensional instability or part distortion over time.
- May reduce fatigue resistance or cause premature failure if not managed.
- Often relieved by annealing after forming.

4.3 EXAMPLES OF PROPERTY MODIFICATION DURING METAL FORMING:

1. Cold Rolling — Increased Strength and Reduced Ductility

- Before: Hot-rolled mild steel has moderate strength and good ductility.
- After: Cold rolling increases yield and tensile strength through work hardening, but the steel becomes less ductile.
- Example: Cold-rolled steel sheets used in car body panels have higher strength but are less formable than hot-rolled sheets.

2. Forging — Grain Refinement for Toughness

- Before: Cast parts have coarse grains, which make them brittle.
- After: Forging compresses and refines the grain structure, increasing toughness and fatigue resistance.
- Example: Forged crankshafts in engines are tough and durable due to their refined grain structure.

3. Drawing — Enhanced Tensile Strength

- Before: Raw metal wire has uniform properties.
- After: During wire drawing, the metal is elongated, causing work hardening and increasing tensile strength, but reducing flexibility.
- Example: Drawn copper wires used in electrical cables have high tensile strength for better performance.

4. Deep Drawing — Induced Anisotropy

- Before: Flat metal sheet has uniform properties.
- After: Deep drawing aligns the grain flow around the drawn shape, making the part anisotropic — meaning its strength and stiffness vary depending on direction.
- Example: Aluminum beverage cans made by deep drawing show directional properties due to anisotropy.

Metal forming doesn't just alter the shape of a part — it directly engineers its mechanical properties. Processes like rolling, forging, and drawing can enhance strength through work hardening, change directional properties via anisotropy, and leave behind internal stresses (residual stresses) that must be managed for long-term part performance.

5. REAL-WORLD APPLICATIONS OF METAL FORMING PROCESSES

1. Automotive Industry — Forging of Engine Components

Application: In the automotive sector, critical engine components like crankshafts, connecting rods, and camshafts are produced through forging.

Process Used: Forging is a process where metal is deformed under high pressure, typically while heated, to achieve the desired shape.

Why This Process is Suitable: Forging creates a very strong and dense grain structure that follows the shape of the part. This enhances the mechanical properties such as fatigue resistance, toughness, and strength — all of which are essential for engine components that undergo repeated cycles of stress and strain. Forged parts outperform cast or machined parts in high-stress applications, which is why the automotive industry prefers this process for internal engine parts.

2. Aerospace Industry — Sheet Metal Forming for Aircraft Fuselage and Wings

Application: Aircraft fuselage panels, wing surfaces, and structural elements are manufactured using various sheet metal forming processes.

Process Used: Sheet Metal Forming methods such as stretching, stamping, and deep drawing are extensively used to create these components.

Why This Process is Suitable: The aerospace industry requires lightweight yet strong components with highly controlled geometries and smooth surface finishes to ensure aerodynamic efficiency. Sheet metal forming not only provides the required dimensional accuracy but also allows the use of advanced alloys like aluminum and titanium, which combine strength with low weight. Moreover, the grain flow and residual stress distribution achieved through this process enhance performance during flight stresses.

3. Construction Industry — Rolling for Structural Steel Beams

Application: Steel beams, columns, and channels used in high-rise buildings, bridges, and industrial structures are produced through rolling.

Process Used: Hot Rolling is commonly used for producing structural steel sections.

Why This Process is Suitable: Rolling is ideal for producing large, uniform cross-sectional shapes like I-beams, H-beams, and channels. The process is fast, economical, and allows the material to be shaped at high temperatures when it is more ductile, making it easier to form into precise geometries. Rolled steel sections also exhibit consistent mechanical properties, which is important for the safety and stability of construction projects.

6. CHALLENGES AND INNOVATIONS:

6.1 CURRENT CHALLENGES IN METAL FORMING PROCESSES:

Metal forming is a crucial process in manufacturing industries such as automotive, aerospace, and construction. Despite technological advancements, several challenges persist that affect efficiency, quality, and sustainability. Here are some of the key challenges:

1. Material Limitations

- **High-Strength Alloys**: The growing demand for stronger, lightweight materials (like high-strength steels, titanium, and superalloys) makes forming more difficult due to their **reduced ductility** and **higher forming forces** required.
- **Brittle Materials**: Certain metals, especially at low temperatures, can become brittle and **crack during deformation**, limiting their use in cold forming processes.
- **Material Variability**: Inconsistent material properties (e.g., thickness, grain structure) can cause defects like wrinkling, tearing, or springback, especially in **sheet metal forming**.

2. Process Optimization

- **Complex Process Parameters**: Controlling variables like temperature, strain rate, and deformation speed is critical. Even small errors can lead to **dimensional inaccuracies** or **internal defects**.
- **Tool and Die Wear**: High pressures and temperatures lead to **rapid tool degradation**, increasing downtime and maintenance costs.
- **Cycle Time vs. Quality Trade-Off**: Faster production increases output but may reduce accuracy or surface finish. Balancing speed with precision remains a big challenge.

3. Cost and Energy Consumption

- **High Initial Investment**: Forging and forming equipment are expensive to acquire and maintain, making it hard for small and medium enterprises (SMEs) to adopt modern forming technologies.
- **Energy-Intensive Processes**: Hot forging and other thermomechanical processes consume large amounts of energy, contributing to **high operational costs** and **carbon emissions**.

4. Lack of Skilled Workforce

- **Labor Shortage**: The forging and metal forming industry requires **highly skilled technicians** to manage machinery and monitor quality, but there is a shortage of trained professionals in many regions.
- **Training Needs**: As digital tools like simulation and automation are introduced, workers need **continuous upskilling** to keep pace with technological advancements.

5. Integration of Digital Technologies

- **Adoption of Industry 4.0**: Technologies like IoT, AI, and digital twins are underutilized in traditional forging setups. Implementing these requires investment, integration, and change management.
- **Simulation Accuracy**: While simulation software helps optimize forming processes, limitations in **material modeling and prediction accuracy** can still lead to unexpected results during production.

6. Sustainability and Environmental Concerns

- **Material Waste**: Processes like closed-die forging can produce excess material (flash), which adds to cost and environmental impact.
- **Emissions and Pollution**: The heating processes in metal forming generate **CO_2 emissions** and require **sustainable alternatives** like green energy or lower-temperature forming.

6.2 RECENT INNOVATIONS AND TECHNOLOGICAL ADVANCEMENTS IN METAL FORMING:

The metal forming industry has seen significant innovation in recent years due to increasing demands for high-performance materials, lightweight structures, and cost-effective, sustainable manufacturing. Below are the **key advancements** transforming how metal forming processes are designed and implemented.

1. Advanced Materials for Forming

High-Strength Lightweight Alloys

- **Aluminum-Lithium alloys, Titanium alloys, and Advanced High-Strength Steels (AHSS)** are now widely used in aerospace and automotive industries.
- These materials allow for **weight reduction** without compromising strength, improving fuel efficiency and performance.

Superplastic Materials

- Certain alloys like **Superplastic Titanium** can undergo **extreme elongation** during forming without cracking.
- Used in **superplastic forming** processes for complex shapes in aerospace components.

2. Computer-Aided Design and Manufacturing (CAD/CAM)

Finite Element Analysis (FEA) in Forming Simulation

- Advanced software tools simulate the metal forming process (deformation, heat transfer, stress distribution), allowing engineers to:

➢ Optimize die design

➢ Minimize defects like wrinkles, cracks, or springback

➢ Reduce trial-and-error during production setup

Digital Twins and Smart Manufacturing

- A **digital twin** is a real-time digital replica of a physical forming process.
- Enables real-time monitoring, predictive maintenance, and adaptive control, improving efficiency and product quality.

3. Artificial Intelligence (AI) and Machine Learning

- AI-driven systems are being integrated into forming lines to:

➢ Predict material behavior
➢ Automate quality control
➢ Optimize process parameters for speed, energy, and material efficiency

- **Example**: AI-based defect detection systems using image recognition to flag inconsistencies in formed parts.

4. Integration with Additive Manufacturing (Hybrid Manufacturing)

- Hybrid processes combine **3D printing (additive)** with **forging or forming (subtractive/deforming)**.
- Example: **Additive-Forging**, where a near-net shape is 3D-printed and then forged to improve grain structure and mechanical properties.
- Reduces material waste and shortens production time for customized components.

5. Warm and Cold Forming of Difficult Materials

- Innovations in **warm forming** enable the shaping of **magnesium and aluminum** at lower temperatures, reducing energy consumption while maintaining formability.
- **Cryogenic forming** is also emerging, where forming is done at very low temperatures to enhance strength and reduce springback.

6. Advanced Tooling and Die Technologies

- **Additive Manufactured Dies** with conformal cooling channels increase die life and improve temperature control during forging.
- **Surface-treated or coated dies** (e.g., with ceramics or nitrides) reduce wear and increase production cycles.

7. Sustainability and Eco-Friendly Innovations

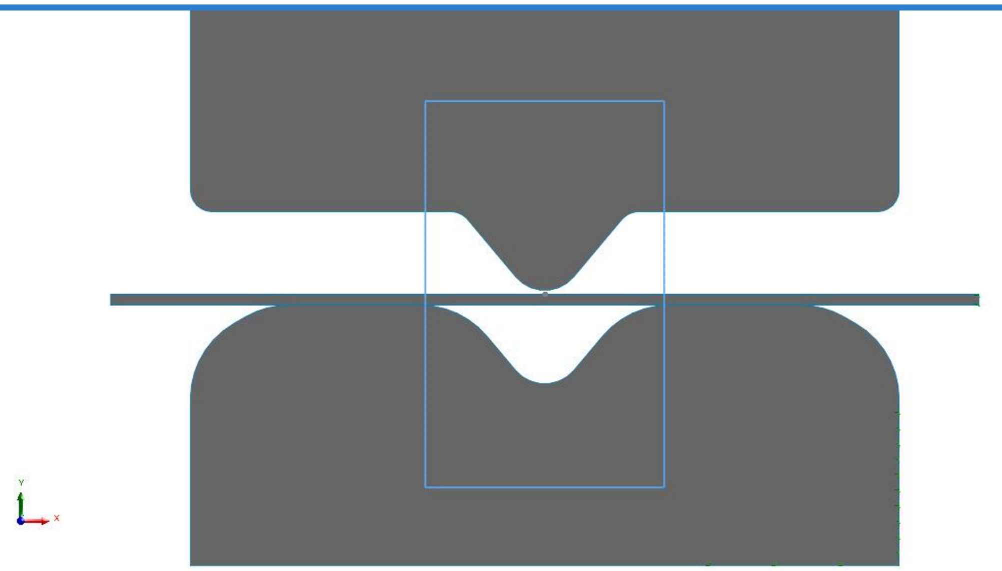
- **Near-net shape forming**: Reduces waste by minimizing machining.
- **Green energy integration**: Using solar, electric induction, or waste heat recovery systems in forging shops.
- **Recyclable lubricants and coolants**: More eco-friendly alternatives replacing traditional oils.

8. Real-World Examples

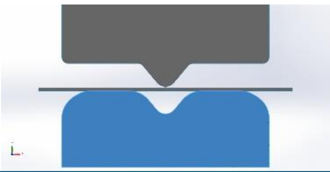
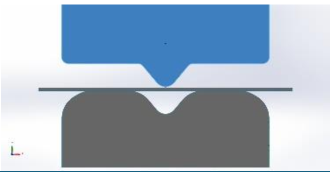
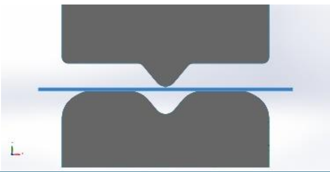
- **BMW**: Uses AI-integrated press lines to optimize metal forming for car body panels.
- **GE Aviation**: Uses hybrid additive + forging methods to produce titanium-aluminide turbine blades.
- **Nippon Steel**: Developed ultra-high-strength, cold-formable steel for lightweight vehicles.

7. SIMULATION:

7.1 NON-LINEAR SIMULATION ON SOLIDWORKS:



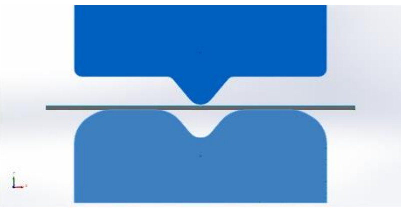
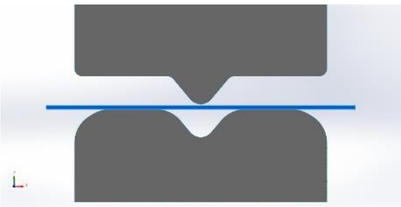
Model name: Sheet Metal Assembly New
Current Configuration: Default

2D Bodies			
Document Name and Reference	Study type	Volumetric Properties	Document Path/Date Modified
Shell-1 	Plane Stress	Section depth:80 mm Weight::223.533 N Volume:0.00292131 m^3 Mass::22.7862 kg Density::7,800 kg/m^3	C:\Users\Hassan Ali\Desktop\Sheet Metal Assembly New.SLDASM Apr 18 03:01:29 2025
Shell-2 	Plane Stress	Section depth:80 mm Weight::209.861 N Volume:0.00274264 m^3 Mass::21.3926 kg Density::7,800 kg/m^3	C:\Users\Hassan Ali\Desktop\Sheet Metal Assembly New.SLDASM Apr 18 03:01:29 2025
Shell-3 	Plane Stress	Section depth:80 mm Weight::4.23792 N Volume:0.00016 m^3 Mass::0.432 kg Density::2,700 kg/m^3	C:\Users\Hassan Ali\Desktop\Sheet Metal Assembly New.SLDASM Apr 18 03:01:29 2025

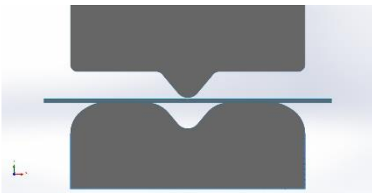
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

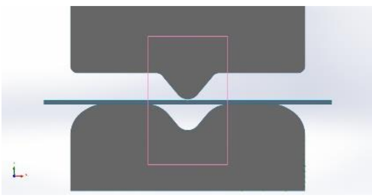
Model Reference	Properties	Components
	Name: **Plain Carbon Steel** Model type: **Linear Elastic Isotropic** Default failure criterion: **Max von Mises Stress** Yield strength: **2.20594e+08 N/m^2** Tensile strength: **3.99826e+08 N/m^2** Elastic modulus: **2.1e+11 N/m^2** Poisson's ratio: **0.28** Mass density: **7,800 kg/m^3** Shear modulus: **7.9e+10 N/m^2** Thermal expansion coefficient: **1.3e-05 /Kelvin**	**SurfaceBody 1(Fillet1)(Die New-1), SurfaceBody 1(Fillet1)(Press New-1)**
Curve Data:N/A		
	Name: **1060 Alloy** Model type: **Linear Elastic Isotropic** Default failure criterion: **Max von Mises Stress** Yield strength: **2.75742e+07 N/m^2** Tensile strength: **6.89356e+07 N/m^2** Elastic modulus: **6.9e+10 N/m^2** Poisson's ratio: **0.33** Mass density: **2,700 kg/m^3** Shear modulus: **2.7e+10 N/m^2** Thermal expansion coefficient: **2.4e-05 /Kelvin**	**SurfaceBody 1(Boss-Extrude1)(Sheet Metal New-1)**
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: **3 edge(s)** Type: **Fixed Geometry**

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-88.2305	7.43169e+06	0	7.43169e+06
Reaction Moment(N.m)	0	0	0	1e-33

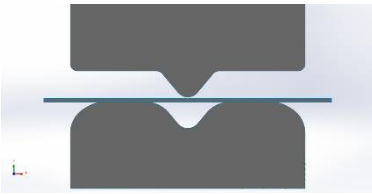
Fixture name	Fixture Image	Fixture Details
Reference Geometry-1		Entities: **1 edge(s), 1 plane(s)** Reference: **Front Plane** Type: **Use reference geometry** Translation: **0, -37, ---** Rotation: **---, ---, ---** Units: **mm, rad**

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	19,012.5	-7.42675e+06	0	7.42677e+06

Fixture name	Fixture Image	Fixture Details

Reaction Moment(N.m)	0	0	0	1e-33

Fixture name	Fixture Image	Fixture Details
Fixed-2		Entities: 2 edge(s) Type: Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-18,978.1	-4,945.17	0	19,611.8
Reaction Moment(N.m)	0	0	0	1e-33

Mesh information

Mesh type	Planar 2D Mesh
Mesher Used:	Blended curvature-based mesh
Maximum element size	10.8125 mm
Minimum element size	10.8125 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	3173
Total Elements	1466
Time to complete mesh(hh;mm;ss):	00:00:02
Computer name:	

Resultant Forces:

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-53.8125	-3.00848	0	53.8965

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

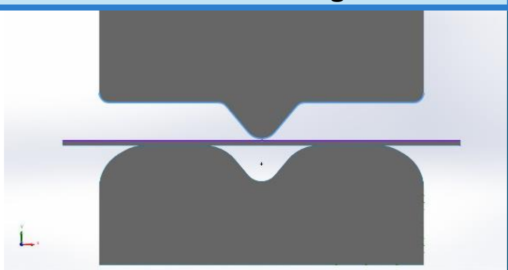
Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0	0	0	0

Free body moments

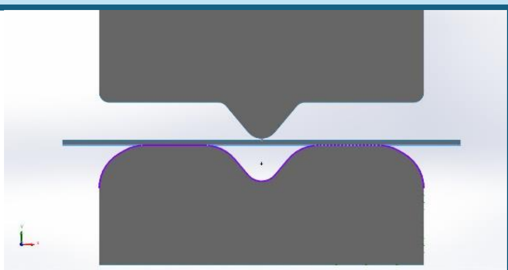
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Interaction Information

Interaction	Interaction Image	Interaction Properties
Local Interaction-1		Type: Contact interaction pair Entities: 10 edge(s) Friction Value: 0.05 Advanced: Surface to surface

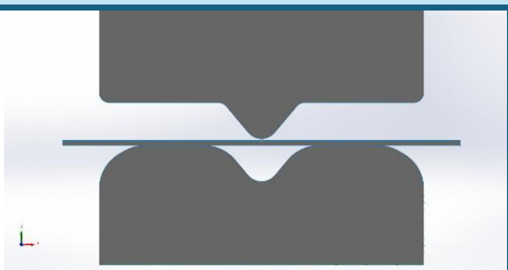
Contact/Friction force

Components	X	Y	Z	Resultant
Contact Force(N)	8.1491E-10	-2.9104E-10	0	8.6532E-10
Friction Force(N)	7.276E-12	-1.4552E-11	0	1.627E-11

Interaction	Interaction Image	Interaction Properties
Local Interaction-2		Type: Contact interaction pair Entities: 14 edge(s) Friction Value: 0.05 Advanced: Surface to surface

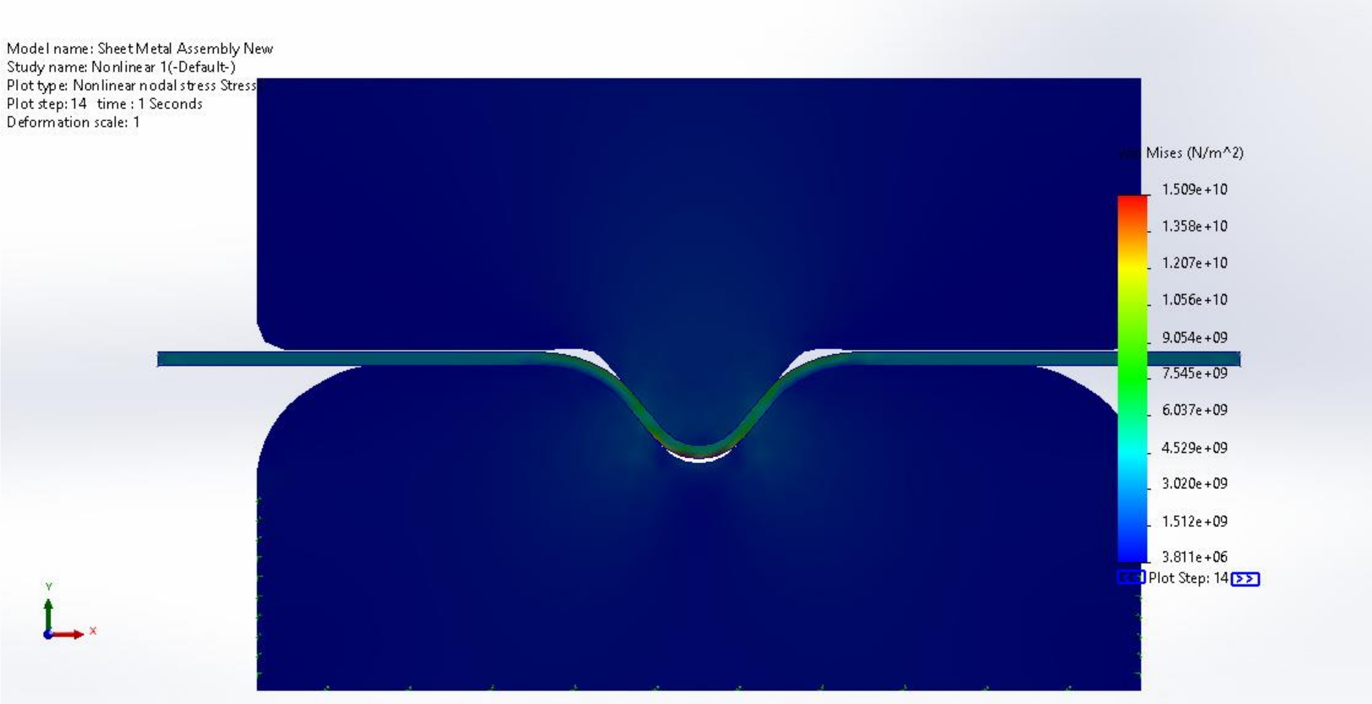
Contact/Friction force

Components	X	Y	Z	Resultant
Contact Force(N)	-8.7311E-11	-1.0768E-09	0	1.0804E-09
Friction Force(N)	1.4552E-11	-1.2369E-10	0	1.2454E-10

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh

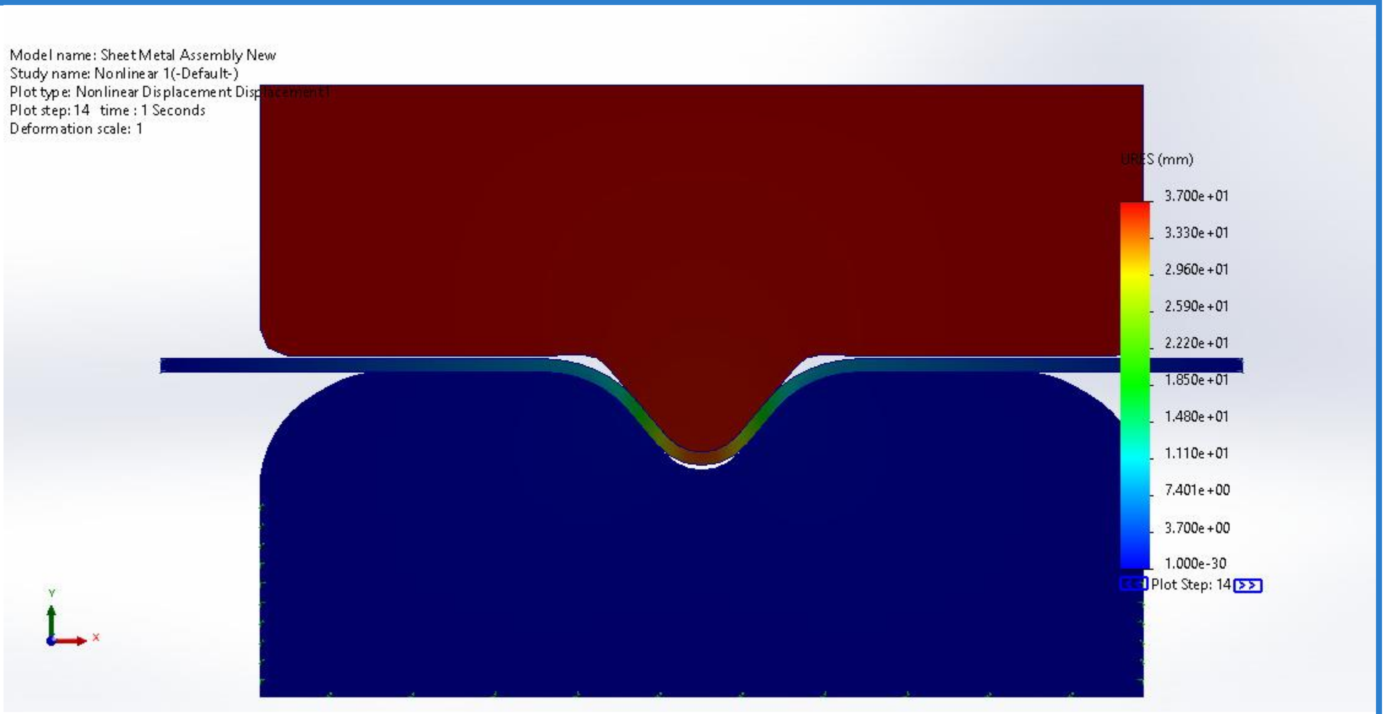
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress at Step No: 14(1 Seconds)	3.811e+06N/m^2 Node: 2264	1.509e+10N/m^2 Node: 2962



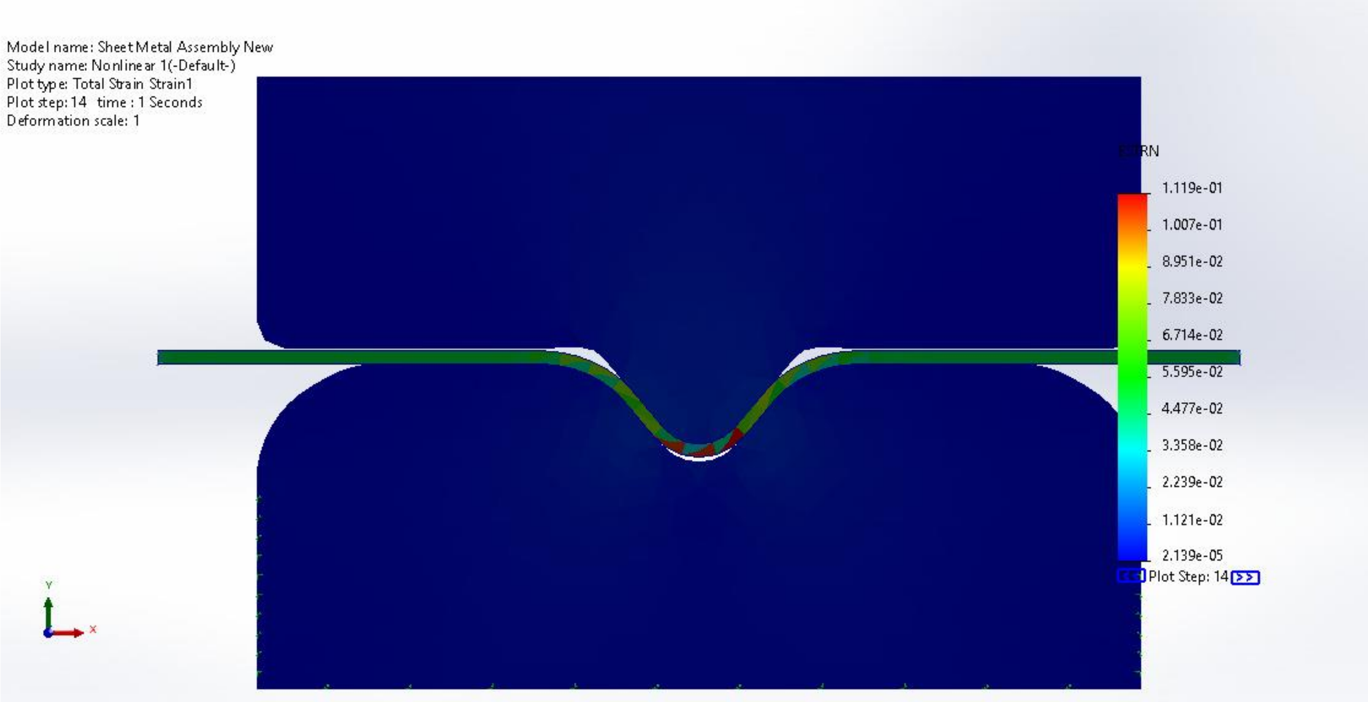
Sheet Metal Assembly New-Nonlinear 1-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement at Step No: 14(1 Seconds)	0.000e+00mm Node: 1	3.700e+01mm Node: 1558



Sheet Metal Assembly New-Nonlinear 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain at Step No: 14(1 Seconds)	2.139e-05 Element: 803	1.119e-01 Element: 1421



Sheet Metal Assembly New-Nonlinear 1-Strain-Strain1

7.2 DOCUMENTATION:

1. Setup

- **Model Name: Sheet Metal Assembly New**
- **Geometry: 2D bodies using planar stress assumption with three shell parts (two carbon steel, one aluminum alloy).**
- **Material Properties:**

➢ ***Plain Carbon Steel:* Yield strength = 220.594 MPa, Elastic modulus = 210 GPa**

➢ ***1060 Aluminum Alloy:* Yield strength = 27.57 MPa, Elastic modulus = 69 GPa**

- **Mesh Details: High-quality, blended curvature-based planar 2D mesh with 3173 nodes and 1466 elements.**

2. Procedure

1. **The model was simplified to 2D using shell elements to reduce complexity.**
2. **Materials were assigned based on typical structural usage (steel for dies, aluminum for sheet metal).**
3. **Fixtures were applied on edges to simulate clamping and support conditions.**
4. **Contacts were defined:**

➢ **Local interactions (surface-to-surface with friction coefficient = 0.05)**

➢ **Global bonded interactions for assembly integrity.**

5. **A force control method was used with a nonlinear solver (Newton-Raphson) and Newmark integration.**

6. **Simulation was run for 1 second with automatic time stepping to capture nonlinear plastic deformation accurately.**
7. **Thermal effects were enabled with a reference (zero strain) temperature of 298 K to consider minor thermal expansion effects.**

3. Observations

- **Contact Behavior: Small but measurable contact and friction forces were generated between surfaces.**
- **Reaction Forces: Fixed supports and reference geometry points reported significant reaction forces (in the order of 7.43 MN).**
- **Deformation Behavior: Maximum displacement reached 37 mm at node 1558.**
- **Stress Distribution: Peak von Mises stress observed: 15.09 GPa (likely localized and possibly indicating stress concentration).**
- **Strain: Max equivalent strain: 0.112, suggesting localized plastic deformation well beyond the elastic limit.**

4. Results

Result Type	Minimum	Maximum	Location
Von Mises Stress	3.81 MPa	15.09 GPa	Node 2962
Displacement	0 mm	37 mm	Node 1558
Strain	2.13e-05	0.112	Element 1421

7.3 Analysis and Theoretical Correlation

- ### Plastic Deformation:
  The simulation clearly enters the plastic region as seen from the equivalent strain and von Mises stress results. This aligns with the theory that metal forming processes (like stamping or deep drawing) cause permanent deformation once stress exceeds the yield strength.
- ### Stress-Strain Behavior:
  High stress concentrations and large deformations are consistent with nonlinear material behavior, where the relationship between stress and strain is no longer proportional. This justifies the use of nonlinear analysis.
- ### Strain Hardening Implication:
  The model likely exhibits strain hardening, a common phenomenon where a material strengthens after plastic deformation. This can be seen in areas with high equivalent strain values.
- ### Anisotropy and Friction Effects:
  Although the simulation assumes isotropic material behavior, real sheet metals often develop anisotropy due to rolling. The frictional contact also hints at possible forming defects like wrinkling or thinning in real scenarios.

8. Conclusion

Metal forming remains a cornerstone of modern manufacturing, enabling the efficient production of strong, lightweight, and complex components across a range of industries including automotive, aerospace, and construction. Through a deep understanding of plastic deformation, stress-strain behavior, and process-specific influences, engineers can manipulate the internal structure and mechanical properties of metals to meet exacting performance requirements.

This assignment has explored the fundamental principles of metal forming, including its classification by stress type and working temperature, and examined key processes such as forging, rolling, extrusion, and drawing. Forging, in particular, was analyzed in detail—highlighting its ability to refine grain structure, enhance toughness, and produce durable parts that outperform those made by casting or machining.

Mechanical concepts like work hardening, anisotropy, and residual stresses were shown to significantly influence final product behavior and reliability. Real-world applications demonstrated how industry-tailored forming methods contribute to structural integrity and product efficiency, from automotive crankshafts to aerospace fuselage panels.

The simulation of the "Sheet Metal Assembly New" model provided valuable insights into how theoretical knowledge is applied in digital environments. The nonlinear analysis accurately captured plastic deformation and stress concentrations, reflecting real-world forming behavior and reinforcing the importance of simulation in modern process design.

Despite its many advantages, metal forming still faces challenges related to material limitations, tool wear, energy consumption, and process control. However, the rise of advanced materials, AI-driven optimization, and sustainable technologies is reshaping the future of forming. Innovations like digital twins, hybrid additive-forming methods, and eco-friendly practices promise to enhance precision, reduce waste, and meet the evolving demands of smart manufacturing.

In conclusion, a strong foundation in metal forming theory, combined with hands-on simulation and awareness of industrial trends, equips engineers with the tools to innovate and excel in the ever-evolving landscape of manufacturing.